

Meridian Retrieval Benchmark, Internal Report R-4471

3. Results

Table 3 reports retrieval quality across six configurations. The evaluation set contains 400 queries, 90 of which contain an exact identifier such as a part number or an error code.

Table 3: Retrieval quality by configuration.

Method	Recall@10	Precision@5	Latency_ms
DENSE_ONLY	0.684	0.512	46
BM25_ONLY	0.601	0.549	11
HYBRID_ALPHA	0.712	0.573	52

HYBRID_RRF	0.831	0.604	58
HYBRID_RRF_RR	0.889	0.702	318
COLBERT_LATE	0.874	0.681	141

HYBRID_RRF fuses the dense and sparse legs with reciprocal rank fusion at $k=60$. HYBRID_RRF_RR adds a cross-encoder reranker over the top 50 candidates, which is where the latency comes from.

Table 4: Configuration parameters.

Parameter	Value	Applies at	Notes
efSearch	128	query time	runtime recall dial
efConstruction	200	build time	no query effect
M	32	build time	edges per node
rrf_k	60	fusion	rank smoothing

Table 5: Legacy results, retained for reference (no ruling lines).

Method	Recall@10	Notes
LEGACY_TFIDF	0.412	superseded by BM25_ONLY
LEGACY_LSA	0.388	superseded by DENSE_ONLY